



Birla Institute of Technology & Science, Pilani

Pilani Campus

ACADEMIC— UNDER GRADUATE STUDIES DIVISION

First Semester 2026-27

Course Handout Part II

Date: 1st August 2026

In addition to part I (General Handout for all courses appended to the Timetable), this portion gives further specific details regarding the course.

Course Number : CE G515

Course Title : Fundamentals of Systems Engineering

Instructor-in-charge : Prof. Rallapalli Srinivas

[Team of Instructors] : Prof. Rallapalli Srinivas

Timings : T Th F (3rd: 10 am to 11 am), 1226

1. **Course description:** Systems Engineering is an interdisciplinary field that focuses on the structured development, integration, and management of complex systems across their entire life cycle—from conceptual design and planning to operation, maintenance, and retirement. In the context of infrastructure, it addresses technical, organizational, and stakeholder-related complexities by employing a holistic, systems-oriented approach. This course introduces students to the fundamentals of systems engineering with a focus on civil and infrastructure systems, preparing them to think critically and act strategically in dynamic, real-world environments.
2. **Scope and Objective:** The course is designed to build a strong foundation in systems engineering methodologies while contextualizing their application to infrastructure planning, development, and management. Students will explore systems architecture, modeling, simulation, optimization, and decision-support frameworks relevant to the life cycle of infrastructure systems. The course also integrates modern tools such as AI, IoT, digital twins, and life-cycle costing techniques to support informed decision-making and innovation. The overarching objective is to develop technically sound, system-aware professionals capable of solving real-world infrastructure problems through interdisciplinary thinking, structured analysis, and evidence-based planning.
3. **Course learning outcomes:**
 - a. Understand systems engineering principles and their application to infrastructure life-cycle management.

- b. Apply systems thinking and modeling techniques to analyze complex infrastructure systems.
- c. Use optimization, simulation, and decision-support tools for infrastructure planning and operations.
- d. Evaluate infrastructure projects using life-cycle costing and basic investment analysis methods.
- e. Introduction to Integrating emerging technologies such as AI and IoT into infrastructure systems design and management.

4. Text books and references:

TB1. Blanchard, B. S., & Fabrycky, W. J. (2010). Systems Engineering and Analysis (5th Edition). Pearson.

TB2. Systems Engineering Fundamentals, Defense Acquisition University Press, 2001 (available on line as pdf, Link: [www.dau.mil/pubs/pdf/SEFGuide 01-01.pdf](http://www.dau.mil/pubs/pdf/SEFGuide%2001-01.pdf))

R1: Hillier F.S. and Lieberman G. J., Introduction to Operations Research, Seventh Edition, TMH, 2001

R2: Hendrickson C. and Au T., Project Management for Construction, Prentice Hall, 2003

R3: Gupta, R. Construction Planning and Technology, CBS Publishers, 2008

R4: Benator B., Thumann A., Project Management and Leadership Skills for Engineering and Construction Projects, The Fairmont Press Inc., Lilburn, GA, USA, 2003

5. **[Method of conduct of the course]:** The course will be conducted through lectures, case discussions, simulation exercises, and group projects. Students will engage with real-world infrastructure problems using analytical tools and modern technologies like AI and IoT. Emphasis will be placed on both conceptual understanding and practical application, with evaluation based on assignments, project work, and exams.

6. Course Plan: weekly schedule for 14 weeks with [3] lectures per week

Week no	Module	Lec No.	Reference	Learning outcome
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1	1. Introduction to basics of System Engineering and related approaches	L1-L3: Overview, applications and major components of systems engineering	T1	Understanding different types of systems and their components
2	2. Solving System Engineering problems using Linear Programming, Queuing theory, and	L4-L7: Overview, Assumption and methods for Solving Linear Programming Problems	R1, R3 R1	Learning application of several techniques of linear programming, and queuing and inventory models for analyzing and solving complex civil engineering systems
3	Inventory control techniques	L8-L10: Basic Structure of Queuing Models, types and applications of models	R1	
4		L11-L13: Introduction to Inventory Control, Type and Applications of Inventory Models		
5	3. Simulation, sampling and forecasting techniques, decision models	L14-L16: Overview, Types of Simulation Models, Generation of Random Numbers, Outline of a major simulation Study L17-L18: Introduction and types of sampling Techniques	R1 Class notes	Identifying the response of a Civil engineering system with respect to the changes in several parameters by developing simulation models. Also, predicting the behavior of the system over time using several forecasting techniques and developing decision models
6				

7		L19-L21: Introduction, Judgmental Forecasting Methods, Time Series Methods	R1	
8		L22-L24: Introduction, Decision making with and without experimentation, decision trees	R1	
9	4. Network scheduling, resource planning and financial management	L25-L28: Introduction, Network, Fundamentals, Critical Path Method, Activity Float and Schedule, Case Problems, Software for Network Scheduling	R2, R3	Learning to effectively control a project's cost and duration with an efficient resource allocation and financial management
10-11		L29-L32: Resource Oriented Scheduling, Scheduling with Resource Constraints and Precedencies	R2, R3	
			R2, R3, R4	
12-13		L33-L37: Life-cycle costing; Financial risk; Public-private partnerships; Cost-benefit and multi-criteria analysis; Techno-economic case studies.		



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13-14	5. Application of AI, IoT, Robotics in Civil Engineering systems	L38-L40: Case studies and discussions	Class notes R1	How to apply cutting edge tools and technologies to solve complex systems
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Suggested Tools and Software

- Modeling & Simulation: AnyLogic, NetLogo, MATLAB/Simulink, Python (SimPy)
- Decision-Making: @RISK, D-Sight, OpenLCA
- Project Management: Primavera P6, MS Project, Navisworks
- AI/IoT: Jupyter, ThingSpeak, NI LabVIEW
- System Modeling: MagicDraw, SysML, Vensim
- Economic Modeling: Excel, Python (numpy_financial), HDM-4, CBA Builder

7. Evaluation Scheme:

Component	Weightage	Duration	Date and time	Comments
[Class discussions/Assignments]	[10%]	[In class]	[In class]	[Open book]
[Handson software usage in class]	[10%]	[In class]	[In class]	[Open book/Open-laptop]
[Project]	[20%]	[3 months]	[In class]	Presentations and reports
[Mid Semester]	[25%]	90 min	10/10 AN2	[Open/close notes]
Comprehensive	[35%]	180 min	16/12 AN	[Open/close notes]

8. **Attendance Policy:** Each student is expected to attend all classes and participate in all evaluation components.
9. **Chamber consultation hours:** Friday 12 noon -1 pm, 4 pm to 5 pm
10. **Makeup Policy:** Only in extreme medical emergency

Prof. Rallapalli Srinivas

Instructor-in-charge

CE G515